PARTH TIWARI

Software Engineer | AI/ML & Full-Stack Development

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SUMMARY

Final-year Computer Science student who builds full-stack and AI/ML systems end to end — LLM-based RAG pipelines, secure authentication, and production-style REST APIs. Completed an AI internship at Google via Medi-Caps University. Comfortable across the stack: React/Redux frontends, FastAPI backends, and applied NLP/LLM integration.

EDUCATION

Medi-Caps University, Indore *2023 – 2027*
B.Tech, Computer Science & Engineering | CGPA: 7.87

EXPERIENCE

Google AI Internship (via Medi-Caps University)

- Completed a structured AI internship program covering applied machine learning and AI development practices.
- Studied the fundamentals of Computer Vision and core Machine Learning concepts, including model training workflows and evaluation basics.
- Gained hands-on exposure to Google's AI and productivity tooling for development and collaboration, including Google Cloud-based AI services and Workspace tools.

PROJECTS

ReadEase — Dyslexia-Assistance Reading Platform
React 18, Redux Toolkit, FastAPI, MongoDB Atlas, JWT (RS256), Google OAuth 2.0, Google Cloud TTS

- Built a full-stack accessibility platform with a FastAPI backend spanning 35+ REST endpoints across 7 routers and a React/Redux frontend.
- Implemented secure JWT (RS256) authentication with Argon2id password hashing and Google OAuth 2.0 login.
- Designed a cascading LLM fallback pipeline (Gemini 2.0 Flash Lite → Gemini 2.0 Flash → Gemini 1.5 Flash → Groq Llama 3.1) for reliable text simplification, validated with BERTScore semantic fidelity checks.
- Integrated Google Cloud Neural2 TTS with SSML and real-time word-level highlighting via timepoint tracking.
- Received an O grade (highest) on the external project viva; developed collaboratively across multiple semesters with a 2-person team.

SCRPQA — RAG-Based Q&A System for the Bhagavad Gita *github.com/BinaryBalls/scrpqa*
React 18, TypeScript, FastAPI, ChromaDB, LaBSE, Groq (LLaMA 3.3 70B)

- Built a full-stack Retrieval-Augmented Generation system over a 700-shloka Sanskrit-English corpus with a custom NLP preprocessing pipeline (sandhi splitting, compound decomposition).
- Implemented a dual-index ChromaDB retrieval layer (Sanskrit + English) with bi-encoder and cross-encoder re-ranking, plus an Adaptive Re-Retrieval feedback loop to improve answer relevance.
- Evaluated retrieval and generation quality using the RAGAS framework; served responses via LLaMA 3.3 70B (Groq) with Google OAuth-secured access.

SKILLS

Languages: Python, JavaScript/TypeScript
Frontend: React, Redux Toolkit, Vite
Backend & Data: FastAPI, MongoDB, ChromaDB, JWT/OAuth 2.0, REST API design
AI/ML: LLM integration (Gemini, Groq/LLaMA), RAG systems, embeddings, prompt engineering, RAGAS/BERTScore evaluation, Computer Vision & ML fundamentals
Core CS: Data Structures & Algorithms, DBMS, Competitive Programming